Bottle Rocket Project

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Objective

Four core parameters make rocket land within 1 meter of a target 92 meters away

- Initial air gauge pressure : $\mathscr{P}^{i}$air
- Initial water volume fraction
- Drag coefficient : C_D
- Initial launch angle : θ^{i}



Methodology

- For rocket to land 92 meters out, we need to work with four parameters

- Test each parameter individually to optimise for farthest flight

- Invoke objective for the rocket to land 92 meters out and adjust

Results

Our distance : 92.28 meters

Optimizing each parameter produced varying impacts :

Drag Coefficient :

- Given constant : 0.425
- Farthest mathematically possible distance at : 0
- Farthest realistically possible distance at : 0.425
- Adjusted for ninety-two meters : 0.35

Initial Gauge Pressure :

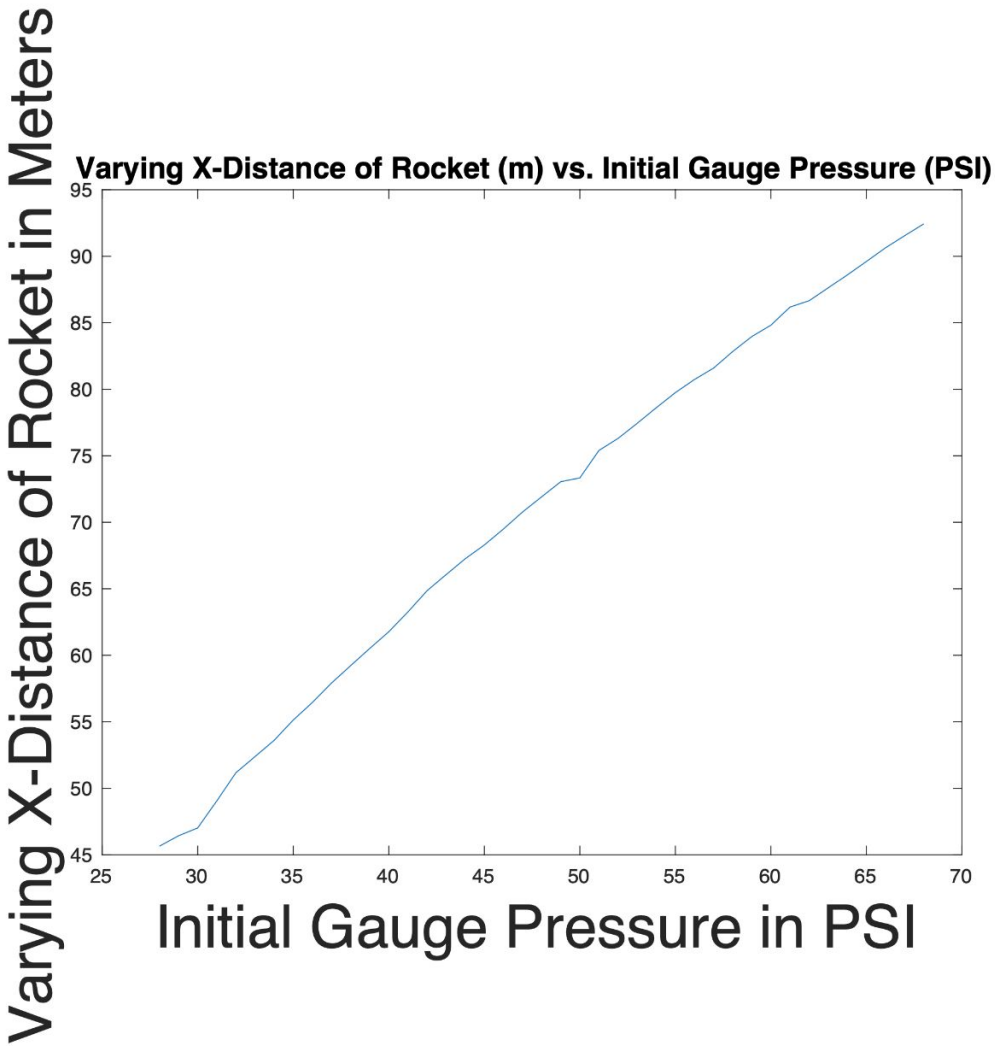
- Given constant : 48 psig
- Farthest possible distance at : 68 psig
- Adjusted for ninety-two meters : 58 psig

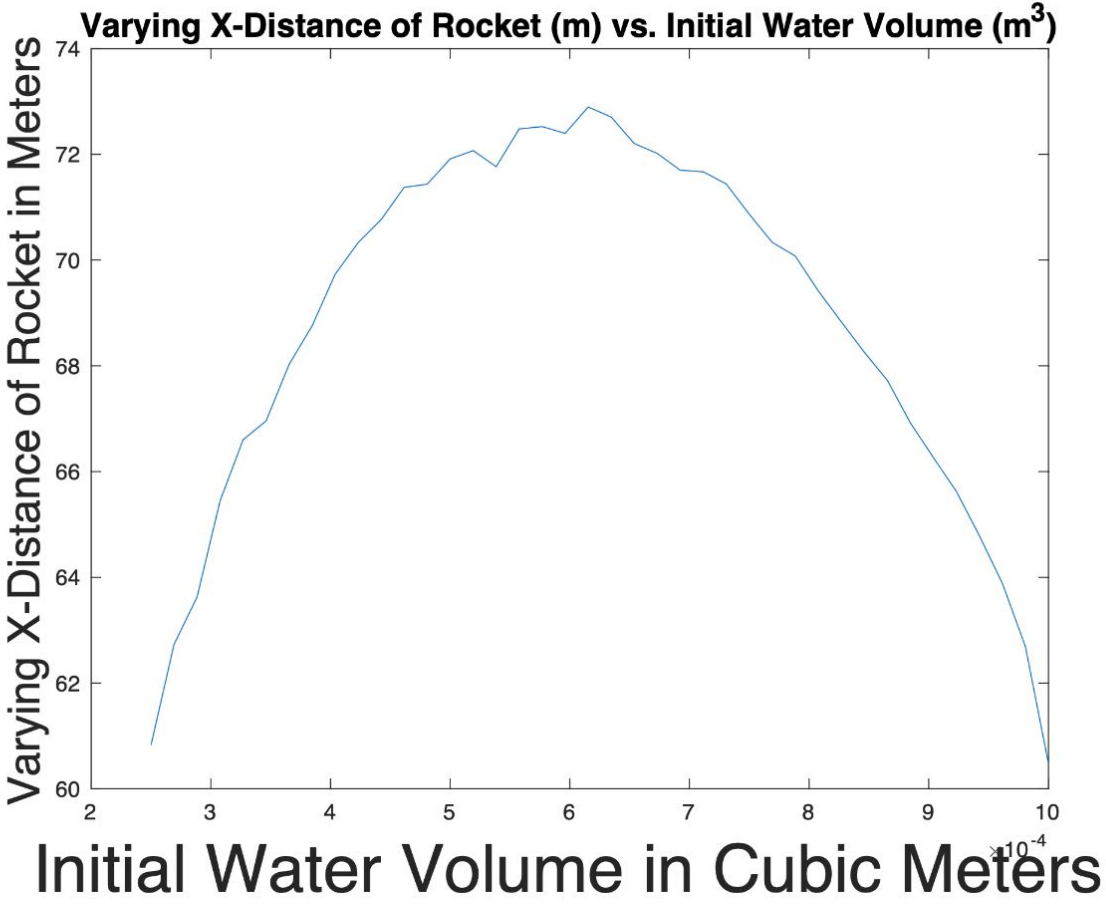
Initial Water Volume :

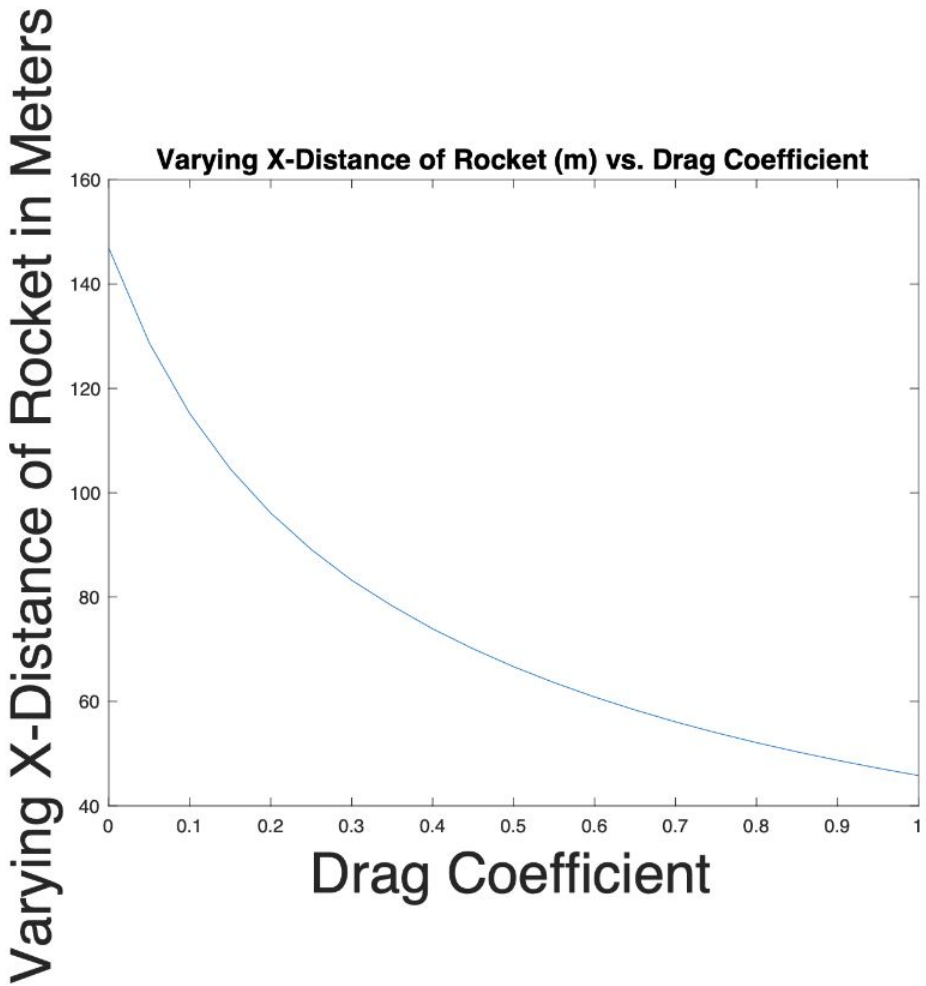
- Given constant : 5×10^{-4} m^3
- Farthest possible distance at : 6.15×10^{-4} m^3
- Adjusted for ninety-two meters : 6×10^{-4} m^3

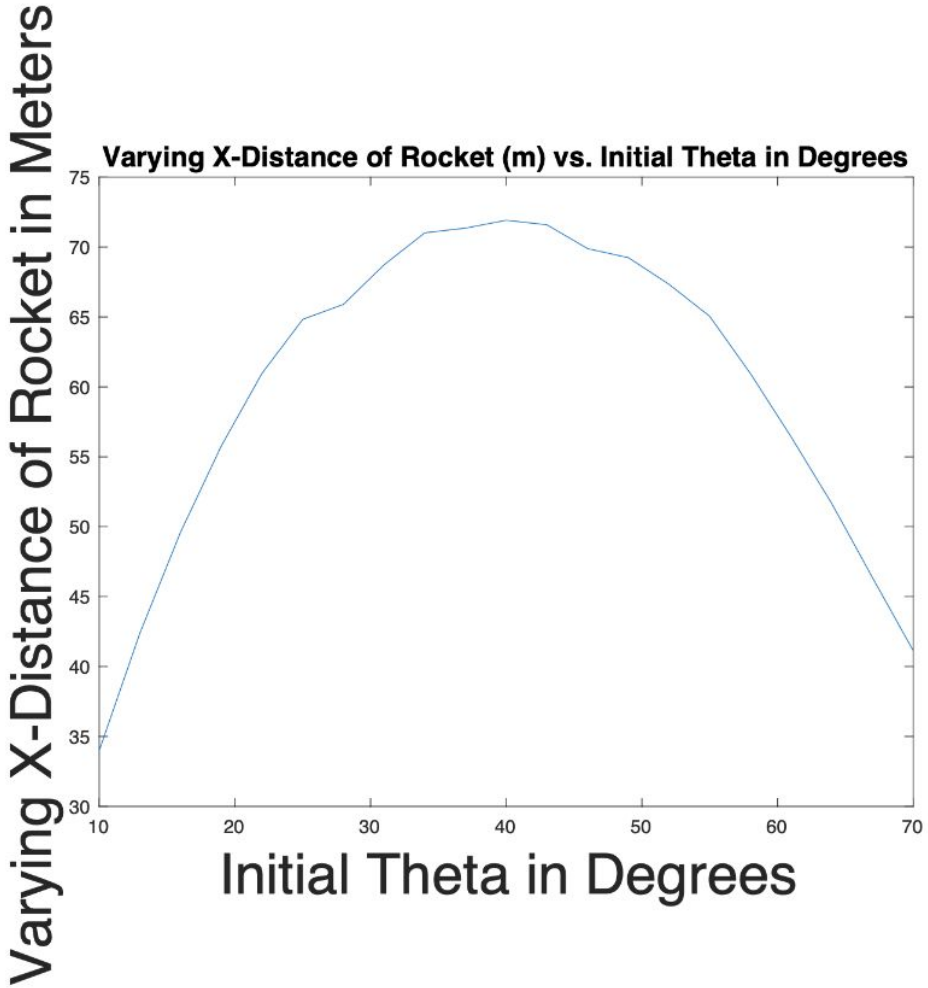
Initial Launch Angle :

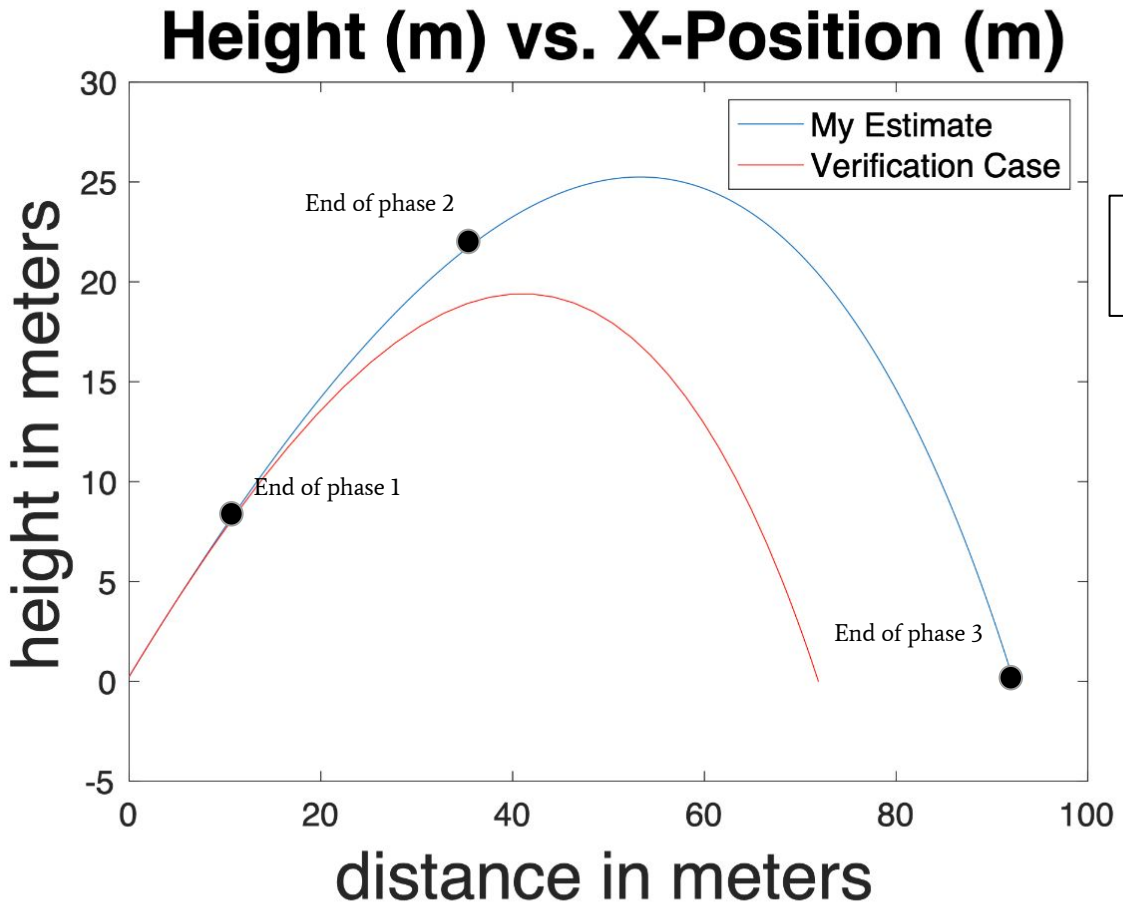
- Given constant : 40°
- Farthest possible distance at : 40°
- Adjusted for ninety-two meters : 40°


Varying X-Distance of Rocket (m) vs. Initial Gauge Pressure (PSI)
Varying X-Distance of Rocket in Meters
95
90
85
80
75
70
65
60
55
50
45
25
30
35
40
45
50
55
60
65
70
Initial Gauge Pressure in PSI

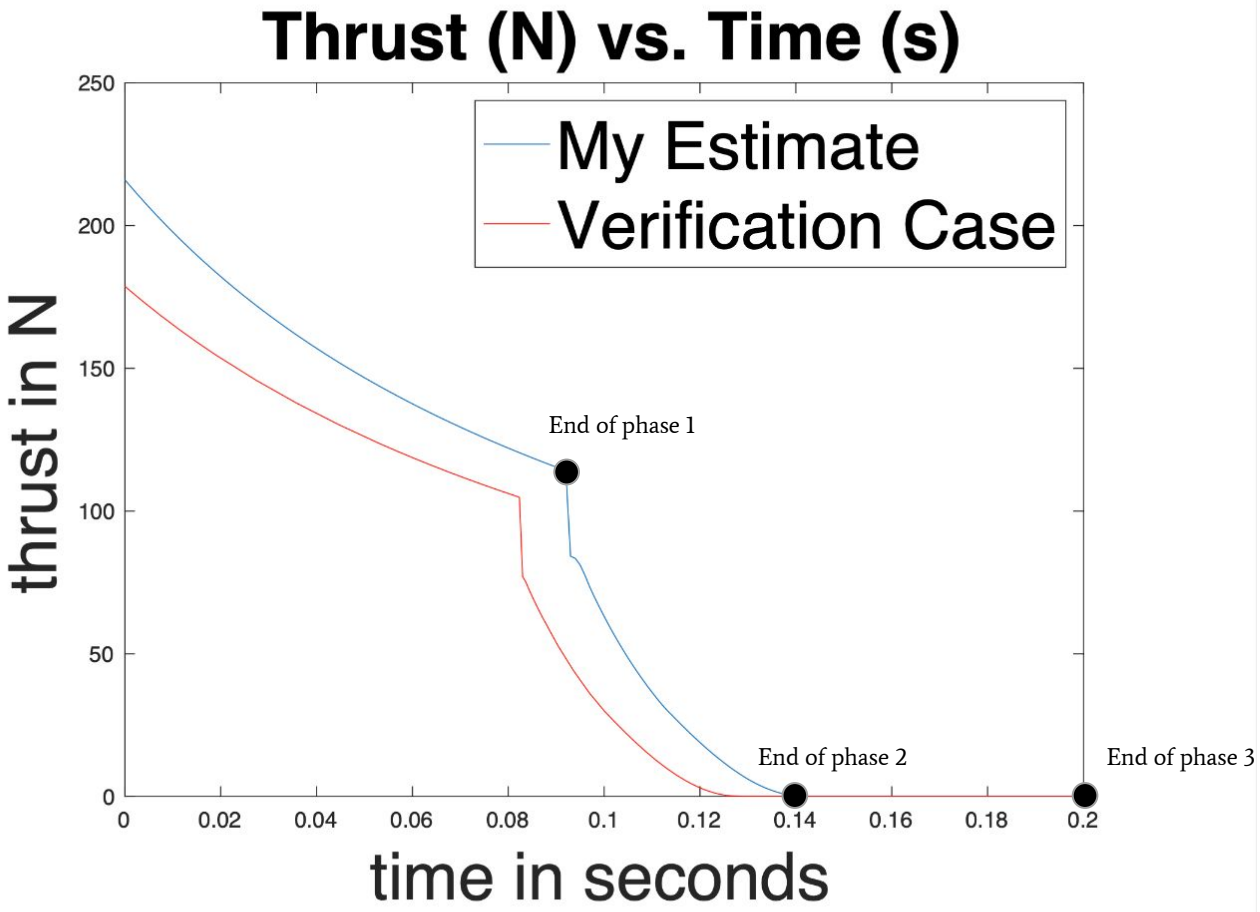

Varying X-Distance of Rocket (m) vs. Initial Water Volume (m^3)
Varying X-Distance of Rocket in Meters
74
72
70
68
66
64
62
60
2
3
4
5
6
7
8
9
10
×10^-4
Initial Water Volume in Cubic Meters


Varying X-Distance of Rocket (m) vs. Drag Coefficient
Varying X-Distance of Rocket in Meters
160
140
120
100
80
60
40
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Drag Coefficient


Varying X-Distance of Rocket in Meters
Varying X-Distance of Rocket (m) vs. Initial Theta in Degrees
75
70
65
60
55
50
45
40
35
30
10
20
30
40
50
60
70
Initial Theta in Degrees



After phase 2, the rest of the flight is ballistic, so the rockets path until the end is phase 3


Thrust (N) vs. Time (s)
My Estimate
Verification Case
thrust in N
time in seconds
End of phase 1
End of phase 2
End of phase 3
250
200
150
100
50
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2

Conclusion

This project showed real-life applications of aerospace engineering

We see how changing parameters drastically affects flight

10-Step Process

1. Purpose of the Problem

The goal is to model the trajectory of a water bottle rocket using numerical integration and analyse how design parameters - initial air gauge pressure, initial water volume, drag coefficient, initial launch angle - affect its flight. The ultimate objective is to adjust these parameters so that the rocket lands within 1 meter of a target located 92 meters away.

2. Collect Information

We collect the constants, given equations, and flight trajectory information.

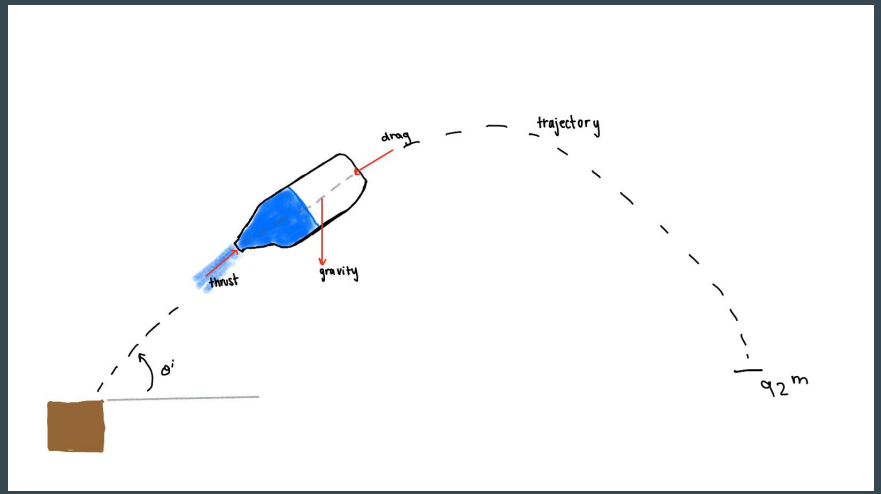
3. Requisite Finds

1. For individual portion, simulate the rocket trajectory to match the verification case (height vs. distance, thrust vs. time).
2. Determine the design parameters for maximum distance
3. Optimise design parameters to achieve a target range of 92 meters with a 1 meter margin.

4. Simplify the Problem by Making Appropriate Assumptions

- Assume the flow is isentropic during air expansion
- Neglect wind effects and assume a two-dimensional trajectory
- Treat water as incompressible and use Bernoulli's equation for flow velocity
- Use constant drag coefficient values
- Assume the rocket does not tumble during flight, maintaining a stable trajectory

5. Sketch



6. Fundamental Principles

- **Newton’s Second Law**: For thrust and drag forces.
- **Bernoulli’s Equation**: To calculate water exhaust velocity.
- **Ideal Gas Law**: To calculate air pressure and density changes.
- **Energy Conservation**: During isentropic air expansion.

7. Alternate Approaches

- Test all possible combinations for our four parameters working together for the farthest possible flight
- Simplify initial models by ignoring drag or assuming constant thrust for validation purposes.
- Optimize parameters manually through trial and error for simple verification scenarios.

8. Step-by-Step Process to Solve the Problem

1. **Model Each Phase**:
   - Phase 1: Use water expulsion equations to calculate thrust, mass flow rate, and air volume.
   - Phase 2: Use air expulsion equations to update thrust and mass flow rate until the gauge pressure equals ambient pressure
   - Phase 3: Simulate ballistic trajectory using gravity and drag.

1. **Numerical Integration**:
   - Use `ode45` to integrate position, velocity, mass, and air volume over time

1. **Verification**:
   - Match simulated results to provide verification data for height, range, and thrust

1. **Optimisation**:
   - Use `ode45` to integrate for the farthest flight
   - Scale the parameters to achieve the 92 meter goal

1. **Validation**:
   - Plot the trajectory, thrust profile, and velocity components

9. Check the Code with Simpler Versions

- Test only with non-zero initial gauge pressure whereby there is no initial water in rocket
- Therefore first phase disappears completely
- Program simplifies significantly

10. Perform a Reality Check

- Ensure the height, range, and thrust values align with real-world expectations
- Check that the optimal design parameters are physically realistic

References

Anderson, J. D., Jr., Introduction to Flight, 7th Ed., McGraw-Hill (2009).

Sutton, G. and Biblarz, O., Rocket Propulsion Elements, 8th Ed., Wiley (2010).

“Maker of MATLAB and Simulink.” *MathWorks*, www.mathworks.com/. Accessed 11 Dec. 2024.